

Reproducibility

Reproducibility I

The artifact is produced by a deterministic, three-step chain that the repository pipeline runs as Stage-02 analysis scripts, in lexicographic order: 00_preflight.py confirms the dependencies import and the edition loads and validates; 10_generate_figures.py writes the halftone scenes and charts to output/figures/; and 20_render_newspaper.py renders the twelve pages to output/pdf/the-triplicate.pdf and emits a machine-readable render report (output/data/render_report.json) plus a human summary. The render is deterministic — the canvas is opened in ReportLab's invariant mode and the figures are drawn from fixed procedural recipes — so the same content yields the same bytes.

Reproducibility I

Correctness is established two ways. A `pytest` suite exercises the pure logic (geometry, configuration, content loaders, typography, figure generation) and renders the real edition end-to-end, asserting that the output is a valid PDF of exactly twelve pages at the correct trim with no over-set page. Because the remaining correctness of a *layout* is visual, the development loop rasterizes each page and inspects it; the project's agent guidance makes that visual check mandatory after any layout change. The engine is type-checked with `mypy` and linted with `ruff`, and the test suite reports roughly ninety-nine percent line coverage.

Every quantitative claim in this project — the page count, the trim dimensions, the figure count — is recorded in `data/claim_ledger.yaml` against the artifact that substantiates it, so a reviewer can trace each number to its source.